

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 1

Mathematics Is Not About Numbers

Assignment

Self-Study Curriculum

Draw a diagram (on paper, not digitally) that shows the major fields of mathematics and what connects them. Do not copy any existing diagram. Use the lecture and the PCM reading as raw material, but the organization must be yours. The diagram should answer the question: “If I am standing in analysis and I need a tool from algebra, what bridge do I cross and what changes when I do?”

Label at least three bridges between fields, and for each one, write one sentence explaining what is preserved and what is lost in the crossing.

Keep this diagram. You will revise it at least twice during this curriculum.